

ZIYUAN ZHANG

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EDUCATION

University of California, Berkeley

Expected May 2029

Doctor of Philosophy in Epidemiology (Infectious Disease track)

- Designated emphasis in Computational Precision Health

Master of Arts in Biostatistics

Harvard University

May 2025

Master of Science in Epidemiology (Infectious Disease track)

Cumulative GPA: 3.96/4.00

University of California, Berkeley

May 2023

Bachelor of Arts in Data Science (Human and Population Health concentration) and Public Health

Cumulative GPA: 3.84/4.00

MANUSCRIPTS

Published

1. **Zhang Z**, Boyer CB, Lipsitch M. Use of the test-negative design to estimate the protective effect of a scalar immune measure: a simulation analysis. *American Journal of Epidemiology*. 2026. <https://doi.org/10.1093/aje/kwag036>
2. Rajwal S, Pandey AK, **Zhang Z**, Chen Y, Liu MX, Das S, Rogers H, Sarker A, Xiao Y. Applications of Natural Language Processing and Large Language Models for Social Determinants of Health: Systematic Review. *Journal of Medical Internet Research*. 2026;28:e83793. <https://doi.org/10.2196/83793>
3. Li G, Puumala E, **Zhang Z**, Rajapakse NS, Ristagno EH, Wolf MJ, Rodning AA, Simner PJ, Gaensbauer JT, Patel R. Cerebrospinal Fluid Metagenomic Next-Generation Sequencing in Pediatric Central Nervous System Infection: Clinical Experience and Diagnostic Stewardship Strategies. *Open Forum Infectious Diseases*. 2026;13(2). <https://doi.org/10.1093/ofid/ofag054>
4. Li G, Hanna C, Dinnes LM, Allen-Rhoades WA, Kohorst MA, Levy ER, Van Dorn CS, **Zhang Z**, Gaensbauer JT. Seraph[®] 100 Microbind[®] utilization in immunocompromised toddler with adenovirus viremia in conjunction with continuous renal replacement therapy and extracorporeal membrane oxygenation. *IDCases*. 2025;42:e02380. <https://doi.org/10.1016/j.idcr.2025.e02380>
5. Li G, Reis CA, Kruc RM, **Zhang Z**, Streck NT, Ristagno EH, Mandrekar J, Wolf MJ, Gaensbauer JT, Patel R. 16S Ribosomal RNA Gene PCR and Sequencing for Pediatric Infection Diagnosis, United States, 2020-2023. *Emerging Infectious Diseases*. 2025;31(13):129-136. <https://doi.org/10.3201/eid3113.241101>
6. **Zhang Z**, Jalali MS, Ghaffar zadegan N. Behavioral Dynamics of Epidemic Trajectories and Vaccination Strategies: A Simulation-based Analysis. *Journal of Artificial Societies and Social Simulation*. 2025;28(1):3. <https://doi.org/10.18564/jasss.5583>
7. Rajwal S, **Zhang Z**, Chen Y, Rogers H, Sarker A, Xiao Y. Applications of Natural Language Processing and Large Language Models for Social Determinants of Health: Protocol for a Systematic Review. *JMIR Research Protocols*. 2025. <https://doi.org/10.2196/66094>
8. Stringfellow EJ, Lim TY, DiGennaro C, **Zhang Z**, Paramasivam P, Bearnot B, Humphreys K, Jalali MS. Long-Term Effects of Increasing Buprenorphine Treatment Seeking, Duration, and Capacity on Opioid Overdose Fatalities: A Model-based Analysis. *Journal of Addiction Medicine*. 2023;17(4):439-446. <https://doi.org/10.1097/ADM.0000000000001153>
9. Stringfellow EJ, Lim TY, Dong H, **Zhang Z**, Jalali MS. The Association Between Longitudinal Trends in Receipt of Buprenorphine for Opioid Use Disorder and Buprenorphine-Waivered Providers in the United States. *Addiction*. 2023;118(11):2215-2219. <https://doi.org/10.1111/add.16291>

In Preparation

10. **Zhang Z**, Keyes KM, Sarker A, Tsai AC, Brown T, Mann JJ, Xiao Y. Trends in Disclosure of Suicidal Intent Among U.S. Suicide Decedents, 2003–2023.

RESEARCH EXPERIENCE

University of California, Berkeley

Sept 2021-Present

Graduate Student Researcher

- Estimating Infection Under-Reporting Fractions from Phylogenetic Trees: Bias Analysis and Validation Aug 2025-Present
 - Supervisor: Joseph A. Lewnard, Ph.D.
 - Developed and applied phylogenetic methods to estimate infection under-reporting, performed bias analyses, validated results with Kaiser and CDPH data, and contributed to manuscript preparation.
- Systematic Review and Meta-analysis of Suicide Prevention Among College Students Before and During the COVID-19 Pandemic Sept 2021-May 2023
 - Supervisors: Timothy T. Brown, Ph.D.; Yunyu Xiao, Ph.D.
 - Conducted the systematic review and meta-analysis; completed title and abstract screening, full text review, and data extraction; assessed literature risks using ROBINS-I and RoB2 tools.
- Literature Review and Meta-analysis of Pediatric Outpatient Infections Attributable to *Streptococcus Pneumoniae* Nov 2022-Jan 2023
 - Supervisor: Joseph A. Lewnard, Ph.D.
 - Conducted literature review and data extraction.

Mayo Clinic

Jan 2024-Present

Division of Pediatric Infectious Diseases

Research Volunteer

Collaborator: Guyu Li, M.D.

- In Vitro Test: Treating *Staphylococcus Epidermidis* Biofilm-Coated Screws with Phage K Dec 2024-Present
 - Designed statistical analysis plan; analyzed and visualized data with Python.
- 16S Ribosomal RNA Gene PCR and Sequencing of Normally Sterile Body Fluids and Tissues for Pediatric Infection Diagnosis, United States, 2020-2023 Mar 2024-Jun 2024
 - Designed statistical analysis plan; analyzed and visualized data with R; drafted and reviewed abstract for conference and manuscript.
- First Use of Seraph® 100 Microbind® Blood Filter in an Immunocompromised Toddler with Viral Viremia on CRRT and ECMO Jan 2024-May 2024
 - Analyzed and visualized data with Python; drafted and reviewed manuscript.

Harvard T.H. Chan School of Public Health

Aug 2023-May 2025

Center for Communicable Disease Dynamics

Graduate Research Assistant

- Assessing the Validity of Test-Negative Designs for Estimating Protection Linked to Exposure-Proximal Correlates of Protection Aug 2023-May 2025
 - Supervisor: Marc Lipsitch, D.Phil.
 - Developed individual-based models, analyzed, and visualized model output with R; drafted and reviewed manuscript.
- Identifying and Analyzing Truly Ambiguous Nucleotide Calls by Position in *Mycobacterium Tuberculosis* Genome Sequencing Sept 2024-May 2025
 - Supervisor: Megan Murray, M.D., Sc.D.; Chuan-Chin Huang, Sc.D.
 - Analyzed the genomic sequence of *Mycobacterium tuberculosis* and validated the low confidence regions in short-read sequencing proposed by existing literature using R.

- Case Studies of Extreme Variations in TB-HIV Death Estimates: Data Source Validation and Assumption Analysis Sept 2024-May 2025
 - Supervisor: Jeffrey Imai-Eaton, Ph.D.
 - Analyzed models employed by WHO and UNAIDS, along with respective data sources and assumptions, to estimate HIV and TB-related HIV prevalence and mortality using R.

Weill Cornell Medicine | NewYork-Presbyterian

Jun 2022-Present

Research Assistant

Supervisor: Yunyu Xiao, Ph.D.; J. John Mann, M.D.

- Applications of Natural Language Processing and Large Language Models for Social Determinants of Health: A Systematic Review Aug 2024-Present
 - Completed title and abstract screening, full text review, and data extraction.
- Trends and Risk Factors in Suicide Deaths with and without Warning From 2003-2022 Jun 2022-Present
 - Cleaned, analyzed, and visualized NVDRS data with R and Joinpoint Trend Analysis Software; drafted and reviewed abstract for conference and manuscript.

Harvard Medical School and MGH Institute for Technology Assessment

Jun 2022-Jan 2025

Research Assistant

Supervisors: Mohammad Jalali, Ph.D.; Navid Ghaffarzadegan, Ph.D.

- Behavioral Dynamics of Epidemic Trajectories and Vaccination Strategies Mar 2023-Jan 2025
 - Developed agent-based models with NetLogo; analyzed and visualized model output with Python; drafted and reviewed manuscript.
- Longitudinal Trends in Buprenorphine Receipt for Opioid Use Disorder and Buprenorphine-Waivered Providers in the United States Aug 2022-Dec 2022
 - Conducted literature review; analyzed and visualized data with Excel; visualized data with Python; drafted and reviewed manuscript.
- Temporal and spatial trends of fentanyl co-occurrence in the illicit drug supply in the United States Aug 2022-Nov 2022
 - Developed pipeline for data preprocessing; created graphical template for state level map combined with bar and line plots; drafted and reviewed abstract for conference.
- Long-term Effects of Increasing Buprenorphine Treatment-Seeking, Duration, and Capacity on Opioid Overdose Fatalities Jun 2022-Oct 2022
 - Cleaned, analyzed, and visualized modeling outputs with R; reviewed manuscript.

Chinese Center for Disease Control and Prevention

Nov 2021-Oct 2022

Center for Global Public Health

Research Intern

Supervisor: Xiaopeng Qi, Ph.D.

- Health Emergency Tabletop Exercise Simulation System
 - Designed tabletop exercise for public health emergency in China; completed literature review; wrote exercise draft with research team for the ASEAN 10+3 National Public Health Emergency Tabletop Exercise Conference.

ORAL PRESENTATION

1. Use of the test-negative design to estimate the protective effect of a scalar immune measure: identification and estimation. *Society for Epidemiologic Research*, June 2025
2. Clinical Experience with 16S Ribosomal RNA Gene PCR Sequencing of Normally Sterile Body Fluids and Tissues in Pediatric Patients. *IDWeek*, October 2024.
3. Trends in Suicide Deaths with and without Warning from 2003-2019: Identifying the Missing Link for Effective Suicide Prevention. *Annual Meeting & Expo, American Public Health Association (APHA)*, November 2022.

AWARD

IDWeek Travel Award for Top Abstracts Submitted in Diagnostics

2024

TEACHING EXPERIENCE

Harvard T.H. Chan School of Public Health

Teaching Fellow: Principles of Clinical Trials

January 2025-March 2025

Tutor: Introduction to Epidemiology; Applied Regression Analysis

September 2024-May 2025

TECHNICAL SKILLS

Adobe Illustrator, Adobe Photoshop, ArcGIS, BEAST2, Covidence, Java, Microsoft Office Suite, NetLogo, Python, R, Tableau